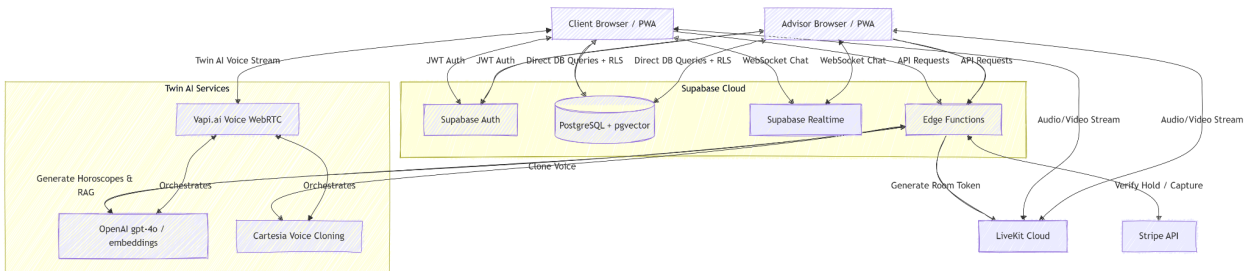


System Architecture Document (SDD)

PROJECT	VERSION	TECH STACK
Cosmically RTC Marketplace	1.0 (Official MVP + Twin AI Spec)	React (Vite) Supabase (PostgreSQL, Realtime, Edge Functions) LiveKit Stripe Vapi.ai

1. High-Level System Context

The platform uses a heavily serverless architecture. The React client communicates directly with Supabase for data and authentication, while relying on secure Edge Functions to broker connections to third-party services (LiveKit, Stripe, and AI providers).



2. Component Architecture

2.1 Frontend (React + Vite)

- **State Management:** React Hooks and Context API for managing user sessions and WebRTC states (`src/hooks/useWebRTC.ts`, `src/hooks/useSessionRealtime.ts`).
- **Routing:** React Router (handling protected routes for `/admin` and `/advisor-portal`).
- **Styling:** Tailwind CSS, utilizing a custom mystical theme palette (`blood-vibrant`, `accent`).

2.2 Backend (Supabase)

- **PostgreSQL Database:** The single source of truth. Protected entirely by Row Level Security (RLS) policies to ensure clients only see their own sessions and transactions.
- **Realtime:** Utilized for the persistent text-chat feature (WhatsApp-style threads) and typing indicators.
- **Edge Functions (Deno):** Secure execution environments for logic that cannot be exposed to the client:
 - **generate-livekit-token:** Issues RTC tokens only after verifying session status and payment holds.
 - **stripe-webhook:** Listens for payment captures and dispute events.
 - **daily-horoscope:** A cron-triggered function that calls OpenAI to populate the **articles** and **horoscopes** tables.

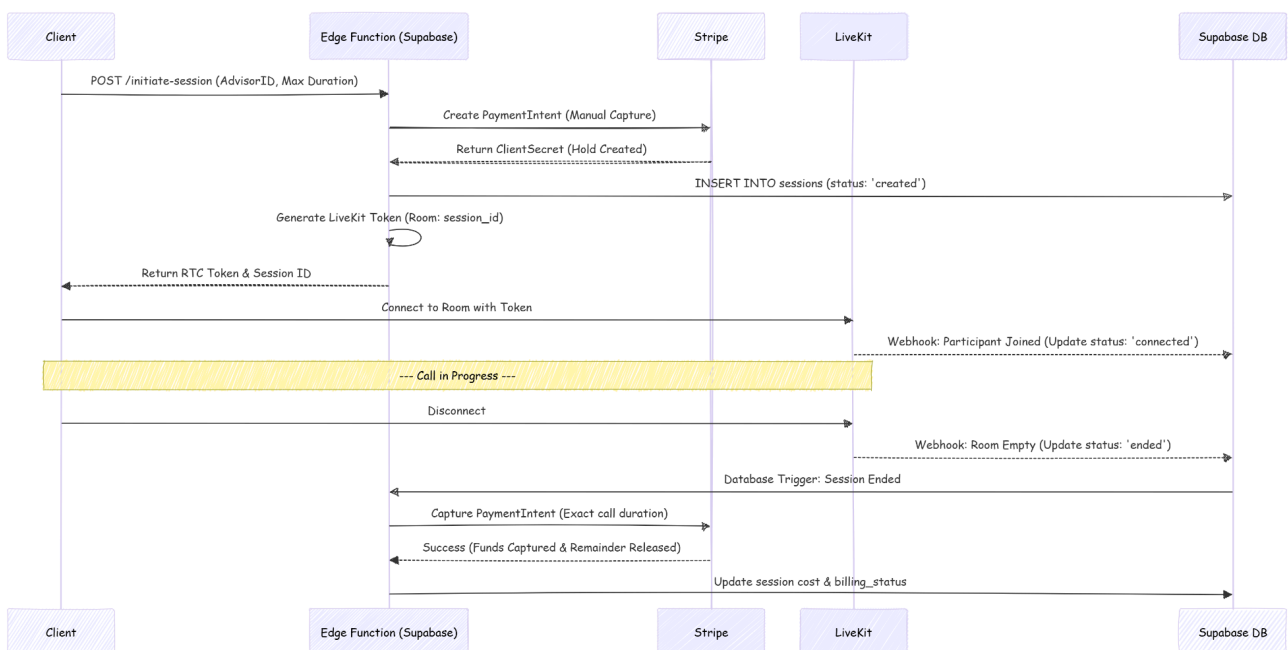
2.3 Real-Time Communications (LiveKit)

- **Topology:** Selective Forwarding Unit (SFU) provided by LiveKit Cloud.
- **Rooms:** Strictly 1:1 topologies. Room names are dynamically generated using the Supabase **session_id** to prevent collisions.

3. Critical System Flows

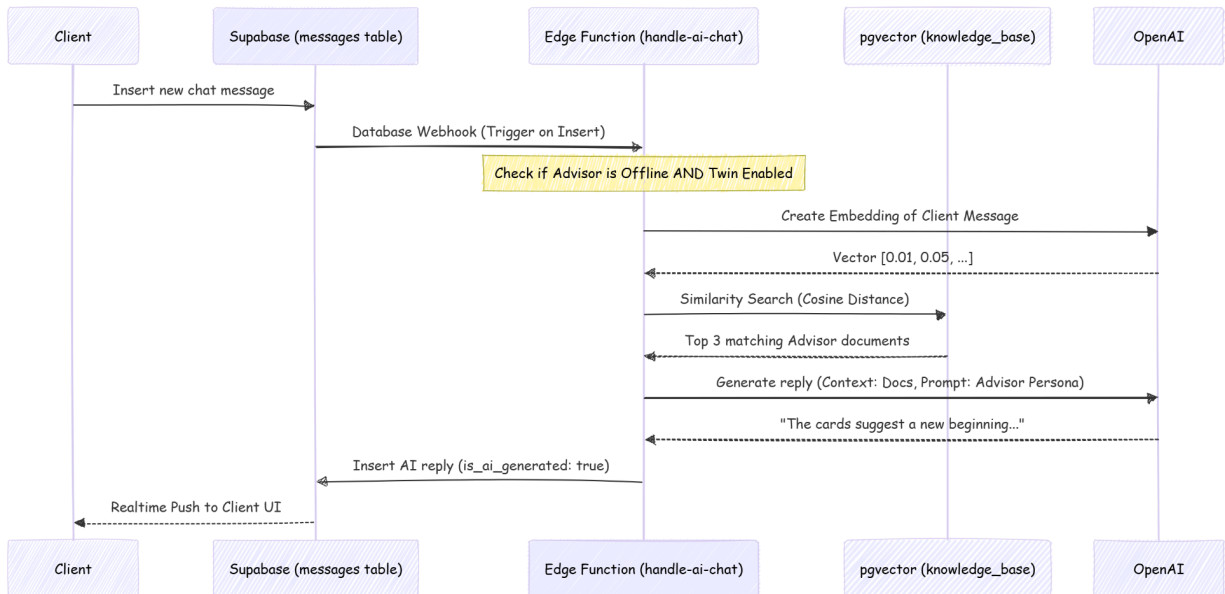
3.1 Session Initiation & Payment (Auth & Capture) Flow

This flow ensures the platform never assumes financial liability for dropped calls while guaranteeing the advisor gets paid for their time.



3.2 Twin AI (Offline Text Mode) Flow

How the system utilizes Retrieval-Augmented Generation (RAG) to reply on behalf of an offline advisor.



4. Security & Compliance

- **Authentication:** Supabase Auth manages JWTs. The frontend passes the JWT in the **Authorization** header to Edge Functions.
- **Row Level Security (RLS):** * *Profiles:* Users can **UPDATE** their own profile. Public can **SELECT** advisor profiles.
 - *Sessions:* **SELECT** and **UPDATE** restricted to `auth.uid() = client_id` OR `auth.uid() = advisor_id`.
- **Secret Management:** No third-party API keys (Stripe Secret, LiveKit Secret, OpenAI Key) are stored in the frontend `.env`. They are strictly managed in the Supabase Vault and accessed only via Edge Functions.
- **Data Deletion:** An automated Edge Function handles GDPR/CCPA "Right to be Forgotten" requests, permanently deleting **profiles** rows and nullifying UUIDs in the **messages** table to preserve chat context without PII.

5. Deployment Infrastructure

Environments

- **Local Development:** Handled via `supabase start` (providing local Postgres, Studio, and Edge Functions) and `npm run dev` for the Vite frontend.
- **Production Frontend:** Currently configured for Vercel deployment (via `vercel.json`). Code is compiled via `npm run build` to static HTML/CSS/JS. Future migration to Dreamhost will involve deploying the `/dist` output folder to the shared/VPS environment.
- **Production Backend:** Hosted on Supabase Cloud (Managed PostgreSQL). Edge functions are deployed directly to the Supabase Deno runtime.

Environment Variables Matrix

VARIABLE NAME	LOCATION	DESCRIPTION
VITE_SUPABASE_URL	Frontend	Supabase project URL
VITE_SUPABASE_ANON_KEY	Frontend	Public routing key for Supabase
LIVEKIT_API_KEY	Supabase Vault	Server-side RTC token generation
LIVEKIT_API_SECRET	Supabase Vault	Server-side RTC security
LIVEKIT_URL	Supabase Vault	WebSocket URL (wss://)
STRIPE_SECRET_KEY	Supabase Vault	Payment Intent generation
STRIPE_WEBHOOK_SECRET	Supabase Vault	Secures incoming Stripe events
OPENAI_API_KEY	Supabase Vault	LLM and Embedding generation
CARTESIA_API_KEY	Supabase Vault	Voice cloning pipeline

DOCUMENT REVISION HISTORY

Version Number	Date	Revisions	Author
1.0	February 19, 2026	-Initial creation	Ivan Pamonag
1.1	February 26, 2026	-Changed ElevenLabs to Cartesia	Ivan Pamonag